

MOODSPACE: AI EMOTIONAL WELLNESS COMPANION

Abstract:

MoodSpace is an AI/ML-powered emotional wellness web application designed to help users express their emotions through text and understand their emotional patterns over time. The system accepts a user's written mood or thought and uses a Natural Language Processing (NLP) based emotion classification system to identify the user's primary emotional state.

The application classifies text into five emotional categories: **Happy, Sad, Angry, Anxious, and Neutral**. In addition to identifying the primary emotion, MoodSpace provides a confidence score, emotion intensity, and, when appropriate, a secondary emotion. The system also stores previous mood entries and uses them to generate a mood journey graph, mood trend prediction, and weekly mood summary.

The backend is implemented using **Python and Flask**, while the emotion classification system uses **TF-IDF vectorization and Multinomial Naive Bayes**, enhanced with a keyword-based emotion lexicon. The system also includes preprocessing techniques for handling negation, contrast clauses, and intensifiers.

Introduction:

Emotions play an important role in a person's daily life. However, users may not always have an easy way to express or understand their emotional state. With the development of Artificial Intelligence and Natural Language Processing, textual information can be analyzed to identify emotional patterns.

MoodSpace was developed as an AI-based emotional wellness companion that allows users to enter their thoughts or feelings in natural language. The system analyzes the input and provides an emotional interpretation along with a supportive response.

The application does not simply perform basic sentiment classification. It combines machine-learning predictions with keyword-based evidence and additional rules for handling situations such as:

- Negated emotions
- Mixed emotions
- Contrast statements
- Emotional intensity
- Ambiguous statements
- Secondary emotions

The detected emotions are then stored for historical analysis, allowing the application to display a user's mood journey and identify whether the recent mood pattern is **Improving, Declining, or Stable**.

Problem Statement:

Traditional mood-tracking applications often depend on users manually selecting their emotional state from predefined options. This can be limiting because people naturally express emotions through sentences rather than fixed labels.

For example:

"I am excited about my presentation tomorrow, but I am also nervous."

This sentence contains more than one emotional state.

Therefore, there is a need for a system that can:

1. Accept natural-language emotional input.
2. Identify the primary emotion.
3. Detect a possible secondary emotion.
4. Estimate confidence in the prediction.
5. Determine emotional intensity.
6. Store historical mood information.

7. Analyze mood patterns over time.
8. Present the information through a simple and user-friendly interface.

MoodSpace addresses these requirements through an NLP and machine-learning based approach.

4. Objectives:

The main objectives of MoodSpace are:

- To develop an AI-based emotional wellness web application.
- To classify user-written text into five emotional categories.
- To improve emotion detection using both ML predictions and keyword evidence.
- To handle negation and contrast expressions.
- To detect secondary emotions in mixed emotional statements.
- To calculate an approximate confidence level for predictions.
- To classify emotional intensity as **Low, Medium, or High**.
- To store users' mood entries for future analysis.
- To visualize historical mood information.
- To predict the recent mood trend.
- To generate a weekly mood summary.
- To provide empathetic responses based on the detected emotion.

Proposed System:

MoodSpace consists of three major components:

1. User Interface

The frontend allows users to enter their thoughts and view:

- User message

- AI response
- Primary emotion
- Confidence
- Intensity
- Secondary emotion
- Mood journey graph
- Mood trend
- Weekly summary
- Recent entries

The interface uses a lavender-themed design with cards, chat-style messages, emotion panels, and responsive layouts.

2. Emotion Analysis Engine

The NLP engine processes the user's text and combines:

TF-IDF + Multinomial Naive Bayes + Keyword Lexicon + Rule-Based Analysis

to determine the emotional state.

3. Mood Analytics

Previous entries are stored in mood_data.csv. The analytics module uses these entries to generate:

- Mood graph
- Mood trend prediction
- Weekly mood summary

The Flask backend connects all these components together.

Technology Stack:

Component	Technology
Programming Language	Python
Web Framework	Flask
Frontend	HTML, CSS, Jinja2
NLP	Scikit-learn
Feature Extraction	TF-IDF
ML Algorithm	Multinomial Naive Bayes
Data Handling	Pandas
Numerical Analysis	NumPy
Visualization	Matplotlib
Data Storage	CSV
Testing	Python test scripts
UI Design	HTML/CSS with responsive layout

The Flask application imports Pandas, Matplotlib and the project's emotion and analytics modules.

Emotion Classification:

8.1 Emotion Categories :-

MoodSpace uses five core emotional categories:

1. Happy
2. Sad
3. Angry
4. Anxious

5. Neutral

These categories are explicitly defined in the classifier.

8.2 Training Dataset :-

The project contains a manually defined training dataset containing example sentences associated with the five emotion categories.

For example:

- "I am feeling really happy today" → Happy
- "I am feeling very sad and down" → Sad
- "I am feeling very mad and furious right now" → Angry
- "I have an exam tomorrow and I am really worried about it" → Anxious
- "Today was just a normal day" → Neutral

The dataset also includes examples involving relief, mixed contexts, negation, and ambiguous expressions.

TF-IDF Feature Extraction:

MoodSpace uses TF-IDF (Term Frequency-Inverse Document Frequency) to convert text into numerical features that can be processed by the machine-learning model.

The implementation uses:

```
TfidfVectorizer  
ngram_range = (1,2)  
stop_words = 'english'
```

This means the classifier considers both individual words and two-word combinations.

For example:

"I am very happy"

can generate features based on words and word pairs such as:

happy

very

very happy

Multinomial Naive Bayes:

After converting the text into TF-IDF features, MoodSpace uses the **Multinomial Naive Bayes** algorithm for emotion classification.

The model is initialized with:

MultinomialNB(alpha=0.3)

The processed training data is transformed using TF-IDF and then used to train the classifier.

The model generates probability scores for each emotion category. These probabilities are then used by the rest of the emotion analysis pipeline.

Keyword-Lexicon Hybrid Approach:

To improve emotion recognition, MoodSpace does not rely only on the machine-learning model.

It also uses an **emotion keyword lexicon**.

For example:

1. Happy

Keywords include:

happy, good, great, excited, love, joy,
cheerful, glad, delighted, relieved

2. Sad

sad, upset, cry, lonely, tired,
hurt, lost, hopeless, disappointed

3. Angry

angry, mad, frustrated, annoyed,
furious, rage, irritated

4. **Anxious**

stress, anxious, worried, nervous,
scared, fear, tense, panic

The keyword evidence is combined with the ML probability scores.

This creates a **hybrid emotion detection system**.

Primary Emotion Detection:

After preprocessing, the system calculates ML probabilities and keyword-based evidence.

The probability scores are combined and normalized.

The emotion with the highest relevant score is selected as the primary emotion.

The system then converts the resulting score into a confidence value, with the implementation restricting the displayed primary confidence to a range of approximately **54%–92%** for non-ambiguous inputs.

Secondary Emotion Detection:

One of the important features of MoodSpace is its ability to identify a **secondary emotion**.

A secondary emotion can be detected when:

- Multiple emotional keywords are present, or
- A conjunction/contrast word is present and the second emotion has sufficient probability.

The implementation considers words such as:

But, also, and, yet, although, however, because, while, though
for mixed emotional expressions.

Emotion Intensity:

MoodSpace also determines the intensity of the detected emotion.

The possible levels are:

- **Low**
- **Medium**
- **High**

The system checks factors such as:

- Primary confidence
- Intensifier words
- Exclamation marks

Mood Data Storage:

Mood entries are stored in:

mood_data.csv

Each entry contains:

Field	Description
date	Date of mood entry
text	User's original input
sentiment	Detected emotion
score	Confidence score

The Flask application automatically creates the CSV structure if the file does not exist or is empty.

After emotion analysis, the detected information is appended to the CSV file.

Mood Journey Visualization:

MoodSpace converts emotions into numerical values:

Emotion	Mood Value
Happy	2
Neutral	1
Sad	0
Angry	-1
Anxious	-2

This allows the system to plot the user's emotional history on a graph.

The graph is titled:

"Your Mood Journey ❤️"

and displays mood level against date.

The provided project output demonstrates this visualization.

Mood Trend Prediction:

The mood_analytics.py module analyzes recent mood entries.

The system requires at least **three valid entries** before generating a trend.

It considers up to the **10 most recent entries** and divides them into an earlier portion and a recent portion. It calculates the average mood level of both portions and compares them.

The result can be:

Improving

If the difference is at least 0.3.

Declining

If the difference is at most -0.3.

Stable

If the difference falls between these values.

User Interface:

The MoodSpace interface is designed with a lavender-themed visual style.

Major UI components include:

Header

MoodSpace 

Your safe space. No judgment. Just understanding.

Chat Section

Users enter their thoughts through a text field and receive responses in a chat-style interface.

Emotion Analysis Card

Displays:

- Primary emotion
- Confidence
- Intensity
- Secondary emotion
- Secondary confidence

Mood Insight

Provides general emotional tracking guidance.

Mood Journey

Displays historical mood information as a graph.

Mood Trend Prediction

Displays:

- Improving
- Declining
- Stable
- Not enough data

Weekly Mood Summary

Displays the numerical mood statistics.

Recent Entries

Displays recent dates and detected moods.

These components are implemented directly in the HTML frontend.

Flask Backend:

Flask acts as the main web framework connecting the frontend and AI components.

The main route:

/

handles both GET and POST requests.

When the user submits a mood:

User Input



`emotion_classifier.analyze()`



Emotion Result



Generate Response



Save to CSV



Generate Graph



Trend Prediction



Weekly Summary



Display Results

The application also includes a /chat endpoint that returns the emotion analysis in JSON format, including primary emotion, confidence, intensity, secondary emotion, and secondary confidence.

Advantages:

1. **Natural-language input**
Users can express their feelings in their own words.
2. **Hybrid AI/ML approach**
Combines machine learning with keyword-based evidence.
3. **Multiple emotion analysis**
Can identify both primary and secondary emotions.
4. **Negation handling**
Attempts to distinguish expressions such as "not angry."
5. **Context handling**
Considers contrast expressions such as "but" and "however."
6. **Emotion intensity**
Provides Low, Medium and High intensity levels.
7. **Historical tracking**
Stores previous mood entries.

8. **Visualization**

Presents mood history using a graph.

9. **Trend analysis**

Identifies improving, declining or stable recent patterns.

10. **User-friendly interface**

Uses a simple, responsive dashboard.

Conclusion:

MoodSpace demonstrates how **Artificial Intelligence, Natural Language Processing, and web technologies** can be combined to create an emotional wellness application.

The system accepts natural-language user input and processes it using a hybrid **TF-IDF + Multinomial Naive Bayes + keyword-lexicon** approach. It identifies one of five core emotional states and provides additional information such as confidence, intensity, and secondary emotion.

Beyond individual emotion classification, MoodSpace stores mood history and uses numerical mood mapping to generate visualizations and recent trend analysis. This makes the project more than a basic text-classification application because it combines **NLP, machine learning, rule-based processing, data storage, visualization, and web development** into one system.

Overall, MoodSpace serves as a practical demonstration of applying AIML techniques to an emotional wellness use case while providing a simple and accessible user interface.

